



ENGINEERING MATHEMATICS - II

Dr. G. K. Jagatheswari

Department of Science and Humanities

ENGINEERING MATHEMATICS - II

UNIT 5 : Fourier series

Session : 6

Sub Topic : Parseval's Identity

Department of Science and Humanities

Parseval's Identity

Parseval's identity relates the squares of the Fourier coefficients to the integral of the square of the function $f(x)$ of period $2l$ defined in the interval $(-l, l)$. That is, if

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right) \dots \dots (1)$$

then,

$$\int_{-l}^l [f(x)]^2 dx = l \left[\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \right]$$

Proof:

Multiplying both sides of (1) by $f(x)$ and integrating term by term from $-l$ to l we get

$$\int_{-l}^l [f(x)]^2 dx$$
$$= \frac{a_0}{2} \int_{-l}^l f(x) dx + \sum_{n=1}^{\infty} \left(a_n \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx + b_n \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx \right) \dots (2)$$

Now,

$$a_0 = \frac{1}{l} \int_{-l}^l f(x) dx \Rightarrow \int_{-l}^l f(x) dx = l a_0$$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx \Rightarrow \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx = l a_n$$

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx \Rightarrow \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx = l b_n$$

Parseval's Identity

Therefore (2) becomes

$$\int_{-l}^l [f(x)]^2 dx = \frac{a_0}{2} (la_0) + \sum_{n=1}^{\infty} (a_n(la_n) + b_n(lb_n))$$

$$\text{Hence } \int_{-l}^l [f(x)]^2 dx = l \left[\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \right]$$

which is the desired Parseval's formula

Parseval's Identity

Sl.No	Period of $f(x)$	Parseval's Identity
1.	$(-l, l)$	$\int_{-l}^l [f(x)]^2 dx = l \left[\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \right]$
2.	$(0, 2l)$	$\int_0^{2l} [f(x)]^2 dx = l \left[\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \right]$

Sl.No	Period of $f(x)$	Parseval's Identity
3.	$(-l, l)$	$2\int_0^l [f(x)]^2 dx = l \left[\frac{a_0^2}{2} + \sum_{n=1}^{\infty} a_n^2 \right]$ (Even Function)
4.	$(-l, l)$	$2\int_0^l [f(x)]^2 dx = l \left[\sum_{n=1}^{\infty} b_n^2 \right]$ (Odd function)

Sl.No	Period of $f(x)$	Parseval's Identity
3.	$(0, l)$	$\underline{2 \int_0^l [f(x)]^2 dx = l \left[\frac{a_0^2}{2} + \sum_{n=1}^{\infty} a_n^2 \right]}$ (Half range cosine series)
4.	$(0, l)$	$2 \int_0^l [f(x)]^2 dx = l \left[\sum_{n=1}^{\infty} b_n^2 \right]$ (Half range sine series)

Obtain the Fourier Series for the function $f(x) = x^2, -\pi < x < \pi$. Hence show that $\sum \frac{1}{n^4} = \frac{\pi^4}{90}$.

Solution:

Here $f(x) = f(-x)$. So x^2 is an even function.

Hence we have

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx, \quad a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx \, dx, \quad b_n = 0$$

and the Fourier expansion of $f(x)$ over $(-\pi, \pi)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

Parseval's Identity - Problems

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx,$$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} x^2 dx,$$

$$= \frac{2}{\pi} \left[\frac{x^3}{3} \right]_0^{\pi}$$

$$a_0 = \frac{2\pi^2}{3}$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx \, dx$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} (x^2) \cos nx \, dx$$

$$= \frac{2}{\pi} \left[(x^2) \left(\frac{\sin nx}{n} \right) - (2x) \left(-\frac{\cos nx}{n^2} \right) + 2 \left(-\frac{\sin nx}{n^3} \right) \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[(2x) \left(\frac{\cos nx}{n^2} \right) \right]_0^{\pi}$$

$$= \frac{2}{\pi n^2} [(2\pi) (\cos n\pi - 0)]$$

$$= \frac{1}{\pi n^2} \cos n\pi (4\pi) =$$

$$a_n = \frac{4}{n^2} (-1)^n$$

Parseval's Identity - Problems

Therefore the required Fourier series is given by,

$$x^2 = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \left(\frac{4}{n^2} (-1)^n \cos nx \right)$$

To deduce, we will use Parseval's identity for even functions.

$$2 \int_0^{\pi} [f(x)]^2 dx = \pi \left[\frac{a_0^2}{2} + \sum_{n=1}^{\infty} a_n^2 \right]$$

$$\text{Therefore, } 2 \int_0^{\pi} [x^2]^2 dx = \pi \left[\frac{1}{2} \frac{4\pi^4}{9} + \sum_{n=1}^{\infty} \left(\frac{4}{n^2} (-1)^n \right)^2 \right]$$

$$2 \left[\frac{x^5}{5} \right]_0^{\pi} = \frac{4\pi^5}{18} + 16\pi \sum_{n=1}^{\infty} \frac{1}{n^4}$$

$$2 \frac{\pi^5}{5} - \frac{2\pi^5}{9} = 16\pi \sum_{n=1}^{\infty} \frac{1}{n^4}$$

$$\Rightarrow \sum \frac{1}{n^4} = \frac{\pi^4}{90}$$



THANK YOU

Dr. G K Jagatheswari

Department of Science & Humanities

jagatheswarigk@pes.edu